

SIMATS ENGINEERING ASSESSMENT TOOL 1

COURSE CODE: CSA51

COURSE NAME: Cryptography and Network Security

COURSE OUTCOME COVERED

CO2: Analyze Public Key crypto system strategies using RSA and key Exchange for Encryption algorithm to strengthen information security. (BL4,BL5)

Analytical Problem Solving: 40%

QUESTIONS:

Total Marks: 50

Annexure A

Code of Conduct:

*I SARU HASINI.S (192521418)_certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

SARU HASINI.S

Annexure A Analytical Problem Solving

1. RSA Encryption Analysis

Given $p=11$, $q=13$

- a. Compute n and $\phi(n)$
- b. Choose $e=7$ and find the private key d
- c. Encrypt the message $M=9$
- d. Decrypt the ciphertext and verify the result
- e. Analyze how changing e affects security

Given:

- $p = 11$
- $q = 13$
- $e = 7$
- Message $M = 9$

(a) Compute n and $\phi(n)$

Step 1: Calculate n

$$n = p \times q = 11 \times 13 = 143$$

Step 2: Calculate Euler's Totient

$$\phi(n) = (p - 1)(q - 1)$$

$$\phi(n) = (11 - 1)(13 - 1) = 10 \times 12 = 120$$

Answer:

- $n = 143$
- $\phi(n) = 120$

(b) Choose $e = 7$ and find the private key d

The private key satisfies:

$$d \times e \equiv 1 \pmod{\phi(n)}$$

So,

$$7d \equiv 1 \pmod{120}$$

Using the Extended Euclidean Algorithm:

$$120 = 17 \times 7 + 1$$

$$1 = 120 - 17 \times 7$$

Therefore,

$$d = -17 \equiv 103 \pmod{120}$$

(Check:)

$$7 \times 103 = 721$$

$$721 \div 120 = 6 \text{ remainder } 1$$

So,

$$\boxed{d = 103}$$

Keys:

- Public key: $(7, 143)$
- Private key: $(103, 143)$

(c) Encrypt the message $M = 9$

RSA Encryption:

$$C = M^e \bmod n$$
$$C = 9^7 \bmod 143$$

Using modular arithmetic:

$$\begin{aligned} 9^2 &= 81 \\ 9^4 &= 81^2 = 6561 \equiv 126 \pmod{143} \\ 9^7 &= 9^4 \times 9^2 \times 9 \\ &= 126 \times 81 \times 9 \\ 126 \times 81 &= 10206 \equiv 53 \pmod{143} \\ 53 \times 9 &= 477 \equiv 48 \pmod{143} \end{aligned}$$

Therefore,

$$\boxed{C = 48}$$

(d) Decrypt the ciphertext

RSA Decryption:

$$M = C^d \bmod n$$
$$M = 48^{103} \bmod 143$$

Using modular exponentiation:

$$48^{103} \bmod 143 = 9$$

Therefore,

$$\boxed{M = 9}$$

The decrypted message matches the original message, so the RSA encryption/decryption process is verified.

(e) Analyze how changing e affects security

The public exponent e affects the efficiency and security characteristics of RSA.

Value of e	Effect
Small e (e.g., 3)	Faster encryption but can be vulnerable to certain attacks if padding is not used correctly.
Common value $e = 65537$	Provides a good balance between security and computational efficiency; widely used in practice.
Large e	Slower encryption and signature verification, with little practical security benefit.

Security considerations:

- e must satisfy:
 - $1 < e < \phi(n)$
 - $\gcd(e, \phi(n)) = 1$
- Choosing a different valid e changes the corresponding private key d .
- The overall security of RSA depends primarily on the difficulty of factoring the modulus n , not on the specific value of e , provided a valid exponent and secure padding scheme are used.
- Very small exponents can introduce vulnerabilities if RSA is implemented without proper padding (such as OAEP for encryption).

2. Block Cipher Mode Evaluation

Given plaintext blocks:

$P = [1010, 1010, 1111, 1010]$, Key = $K = 1100$

- Encrypt using ECB and CBC modes (Assume $IV = 0000$)
- Compare the ciphertext outputs
- Analyze which mode hides patterns better and explain why

Given:

- Plaintext blocks:

$$P = [1010, 1010, 1111, 1010]$$

- Key:

$$K = 1100$$

- IV (for CBC):

$$IV = 0000$$

(a) Encrypt using ECB and CBC modes

1. ECB (Electronic Codebook)

Formula:

$$C_i = P_i \oplus K$$

Plaintext (P_i) Key (K) Ciphertext (C_i)

1010	1100	0110
1010	1100	0110
1111	1100	0011
1010	1100	0110

ECB Ciphertext:

$$[0110, 0110, 0011, 0110]$$

2. CBC (Cipher Block Chaining)

Formula:

$$C_i = (P_i \oplus C_{i-1}) \oplus K$$

where $C_0 = IV = 0000$.

Block 1

$$\begin{aligned} P_1 \oplus IV &= 1010 \oplus 0000 = 1010 \\ C_1 &= 1010 \oplus 1100 = 0110 \end{aligned}$$

Block 2

$$\begin{aligned} P_2 \oplus C_1 &= 1010 \oplus 0110 = 1100 \\ C_2 &= 1100 \oplus 1100 = 0000 \end{aligned}$$

Block 3

$$\begin{aligned} 1111 \oplus 0000 &= 1111 \\ 1111 \oplus 1100 &= 0011 \end{aligned}$$

So,

$$C_3 = 0011$$

Block 4

$$1010 \oplus 0011 = 1001$$

$$1001 \oplus 1100 = 0101$$

So,

$$C_4 = 0101$$

CBC Ciphertext:

[0110, , 0000, 0011 0101]

(b) Compare the ciphertext outputs**Block Plaintext ECB CBC**

1	1010	0110 0110
2	1010	0110 0000
3	1111	0011 0011
4	1010	0110 0101

Observation:

- In **ECB**, identical plaintext blocks produce identical ciphertext blocks.
- In **CBC**, identical plaintext blocks produce different ciphertext blocks because each block depends on the previous ciphertext.

(c) Which mode hides patterns better? Explain.**CBC mode hides patterns better than ECB mode.****Reasons:**

- **ECB Mode**
 - Encrypts each block independently.
 - Identical plaintext blocks always produce identical ciphertext blocks.
 - Repeated patterns remain visible, making ECB less secure for structured data such as images or documents.
- **CBC Mode**
 - Each plaintext block is XORed with the previous ciphertext block before encryption.
 - Even if plaintext blocks are identical, their ciphertext blocks will usually be different.
 - This chaining effect hides repeated patterns and provides better confidentiality.

Conclusion

- **ECB:** Simple and fast, but reveals repeated patterns.
- **CBC:** More secure because it hides plaintext patterns by chaining blocks together with the previous ciphertext and an initialization vector (IV).

3. DES vs 3-DES vs Blowfish Performance Study

Given:

- a. DES key size = 56 bits, time = 2 ms/block
- b. 3-DES key size = 168 bits, time = 6 ms/block
- c. Blowfish key size = 128 bits, time = 1 ms/block

For 1000 blocks of data:

- i. Calculate total encryption time for each algorithm
- ii. Analyze trade-offs between security and performance
- iii. Recommend the best algorithm for secure real-time communication

Given:

- **DES:** Key size = **56 bits**, Time = **2 ms/block**
- **3-DES:** Key size = **168 bits**, Time = **6 ms/block**
- **Blowfish:** Key size = **128 bits**, Time = **1 ms/block**
- **Data size: 1000 blocks**

(i) Calculate total encryption time

Formula:

$$\text{Total Time} = \text{Time per Block} \times \text{Number of Blocks}$$

DES

$$2 \text{ ms} \times 1000 = 2000 \text{ ms}$$

$2000 \text{ ms} = 2 \text{ seconds}$

3-DES

$$6 \text{ ms} \times 1000 = 6000 \text{ ms}$$

$6000 \text{ ms} = 6 \text{ seconds}$

Blowfish

$$1 \text{ ms} \times 1000 = 1000 \text{ ms}$$

$1000 \text{ ms} = 1 \text{ second}$

Summary Table

Algorithm	Key Size	Time/Block	Total Time (1000 Blocks)
DES	56 bits	2 ms	2000 ms (2 s)
3-DES	168 bits	6 ms	6000 ms (6 s)
Blowfish	128 bits	1 ms	1000 ms (1 s)

(ii) Analyze the trade-offs between security and performance

Algorithm	Security	Performance	Analysis
DES	Low	Moderate	Uses a 56-bit key, which is vulnerable to brute-force attacks and is no longer considered

Algorithm	Security	Performance	Analysis
3-DES	High (compared to DES)	Slow	secure. Applies DES three times, increasing security but making encryption about three times slower than DES. It has also been deprecated for many new applications. Uses a variable key size (up to 448 bits). With a 128-bit key, it provides strong security and faster encryption than DES and 3-DES.
Blowfish	High	Fast	

Comparison

- **DES**
 - ✓ Faster than 3-DES
 - ✗ Weak security due to its small 56-bit key
- **3-DES**
 - ✓ Much stronger than DES
 - ✗ Slowest algorithm because encryption is performed three times
- **Blowfish**
 - ✓ Strong security with flexible key sizes
 - ✓ Fastest encryption among the three
 - ✓ Suitable for software implementations and many real-time applications

(iii) Recommend the best algorithm for secure real-time communication

Recommended Algorithm: Blowfish

Reasons:

- **Fastest encryption time:** 1 second for 1000 blocks
- **Strong security:** 128-bit key (and supports even larger keys)
- **Better performance** than DES and 3-DES
- **Efficient** for applications requiring low latency and high throughput

4. Diffie-Hellman Key Exchange Analysis

Given:

- $p=23, g=5$
- User A private key = 6
- User B private key = 15
- Compute public keys for A and B
- Compute the shared secret key
- Demonstrate how a Man-in-the-Middle attacker could alter the exchange
- Suggest a secure method to prevent the attack

Given:

- Prime number: $p = 23$
- Generator: $g = 5$
- User A private key: $a = 6$
- User B private key: $b = 15$

(d) Compute the public keys for A and B

The public key is computed as:

$$\text{Public Key} = g^{\text{Private Key}} \pmod{p}$$

User A

$$A = 5^6 \pmod{23}$$

Calculate:

$$\begin{aligned} 5^2 &= 25 \equiv 2 \pmod{23} \\ 5^4 &= 2^2 = 4 \\ 5^6 &= 5^4 \times 5^2 = 4 \times 2 = 8 \end{aligned}$$

Therefore,

$$\boxed{A = 8}$$

User B

$$B = 5^{15} \pmod{23}$$

Using repeated squaring:

$$\begin{aligned} 5^2 &\equiv 2 \\ 5^4 &\equiv 4 \\ 5^8 &= 4^2 = 16 \end{aligned}$$

Since

$$\begin{aligned} 15 &= 8 + 4 + 2 + 1 \\ 5^{15} &= 5^8 \times 5^4 \times 5^2 \times 5 \\ &= 16 \times 4 \times 2 \times 5 \\ 16 \times 4 &= 64 \equiv 18 \pmod{23} \\ 18 \times 2 &= 36 \equiv 13 \\ 13 \times 5 &= 65 \equiv 19 \end{aligned}$$

Therefore,

$$\boxed{B = 19}$$

(e) Compute the shared secret key

User A computes

$$K = B^a p$$
$$K = 19^6 23$$

Calculate:

$$19^2 = 361 \equiv 16$$
$$19^4 = 16^2 = 256 \equiv 3$$
$$19^6 = 19^4 \times 19^2$$
$$= 3 \times 16 = 48 \equiv 2$$

Thus,

$$\boxed{K = 2}$$

User B computes

$$K = A^b p$$
$$K = 8^{15} 23$$

Using repeated squaring:

$$8^2 = 64 \equiv 18$$
$$8^4 = 18^2 = 324 \equiv 2$$
$$8^8 = 2^2 = 4$$

Since

$$15 = 8 + 4 + 2 + 1$$
$$8^{15} = 4 \times 2 \times 18 \times 8$$
$$4 \times 2 = 8$$
$$8 \times 18 = 144 \equiv 6$$
$$6 \times 8 = 48 \equiv 2$$

Therefore,

$$\boxed{K = 2}$$

Both users obtain the same shared secret.

(f) Demonstrate a Man-in-the-Middle (MITM) attack

Suppose an attacker **Eve** intercepts the communication.

1. A sends public key **8** to B.
2. Eve intercepts it and sends her own public key (for example, **10**) to B instead.
3. B sends public key **19** to A.
4. Eve intercepts it and sends another public key (for example, **7**) to A.

Now:

- **A** computes a secret with Eve (using 7).
- **B** computes a secret with Eve (using 10).
- Eve computes both shared secrets because she knows her own private key(s).

As a result:

- A believes she shares a secret with B.
- B believes he shares a secret with A.
- In reality, both share secrets only with Eve.

Eve can then decrypt, modify, and re-encrypt every message between A and B without either user noticing.

(g) Suggest a secure method to prevent the attack

A **Man-in-the-Middle attack** is prevented by **authenticating the exchanged public keys**. Common methods include:

- **Digital Signatures:** Each party signs its Diffie–Hellman public key with its private signing key, allowing the other party to verify its authenticity.
- **Digital Certificates (PKI):** Certificates issued by a trusted Certificate Authority (CA) verify the identity of each participant.
- **Authenticated Key Exchange Protocols:** Protocols such as **TLS** combine Diffie–Hellman with authentication to prevent MITM attacks.
- **Pre-shared Public Keys or Fingerprint Verification:** Suitable for environments where participants can verify keys out of band.

5. Diffie-Hellman Key Exchange Analysis

Given:

- a. $p=23, g=5$
- b. User A private key = 6
- c. User B private key = 15
- d. Compute public keys for A and B
- e. Compute the shared secret key
- f. Demonstrate how a Man-in-the-Middle attacker could alter the exchange
- g. Suggest a secure method to prevent the attack

Given

- Prime number: $p = 23$
- Generator: $g = 5$
- User A's private key: $a = 6$
- User B's private key: $b = 15$

(d) Compute the public keys for A and B

The public key is calculated as:

$$\text{Public Key} = g^{\text{Private Key}} p$$

User A

$$A = 5^6 23$$

Calculations:

$$5^2 = 25 \equiv 2 \pmod{23}$$

$$5^4 = 2^2 = 4$$

$$5^6 = 5^4 \times 5^2 = 4 \times 2 = 8$$

Public Key of A:

$$\boxed{A = 8}$$

User B

$$B = 5^{15} 23$$

Using repeated squaring:

$$5^2 = 2, 5^4 = 4, 5^8 = 16$$

Since

$$\begin{aligned}15 &= 8 + 4 + 2 + 1 \\5^{15} &= 5^8 \times 5^4 \times 5^2 \times 5 \\&= 16 \times 4 \times 2 \times 5 \\16 \times 4 &= 64 \equiv 18 \\18 \times 2 &= 36 \equiv 13 \\13 \times 5 &= 65 \equiv 19\end{aligned}$$

Public Key of B:

$$\boxed{B = 19}$$

(e) Compute the shared secret key

User A computes

$$\begin{aligned}K &= B^a p \\K &= 19^6 23\end{aligned}$$

Calculations:

$$\begin{aligned}19^2 &= 361 \equiv 16 \\19^4 &= 16^2 = 256 \equiv 3 \\19^6 &= 19^4 \times 19^2 = 3 \times 16 = 48 \equiv 2 \\ \boxed{K} &= \boxed{2}\end{aligned}$$

User B computes

$$\begin{aligned}K &= A^b p \\K &= 8^{15} 23\end{aligned}$$

Calculations:

$$\begin{aligned}8^2 &= 64 \equiv 18 \\8^4 &= 18^2 = 324 \equiv 2 \\8^8 &= 2^2 = 4\end{aligned}$$

Since

$$\begin{aligned}15 &= 8 + 4 + 2 + 1 \\8^{15} &= 8^8 \times 8^4 \times 8^2 \times 8 \\&= 4 \times 2 \times 18 \times 8 \\4 \times 2 &= 8 \\8 \times 18 &= 144 \equiv 6 \\6 \times 8 &= 48 \equiv 2 \\ \boxed{K} &= \boxed{2}\end{aligned}$$

Thus, both users obtain the same shared secret.

Shared Secret Key:

$$\boxed{2}$$

(f) Demonstrate a Man-in-the-Middle (MITM) attack

Suppose an attacker **Eve** intercepts the communication.

1. User A sends public key **8** to User B.
2. Eve intercepts it and replaces it with her own public key (e.g., **10**) before forwarding it to B.
3. User B sends public key **19** to User A.

4. Eve intercepts it and replaces it with another public key (e.g., 7) before forwarding it to A.

As a result:

- User A computes a shared secret with Eve.
- User B computes a different shared secret with Eve.
- Eve knows both secrets and can decrypt, read, modify, and re-encrypt all messages exchanged between A and B.

Neither A nor B realizes that Eve is intercepting their communication.

(g) Suggest a secure method to prevent the attack

A Man-in-the-Middle attack can be prevented by **authenticating the public keys** exchanged during the Diffie–Hellman process.

Common methods include:

- **Digital Signatures:** Users sign their public keys so the recipient can verify their authenticity.
- **Digital Certificates (PKI):** Certificates issued by a trusted Certificate Authority (CA) confirm the identity of each participant.
- **Authenticated Key Exchange Protocols (e.g., TLS):** Combine Diffie–Hellman with authentication to prevent key substitution attacks.
- **Pre-shared Public Keys or Fingerprint Verification:** Useful in environments where users can verify keys through a trusted channel.

6. ECC vs RSA Comparison

Given:

- RSA key size = 1024 bits
- ECC key size = 160 bits (equivalent security)
- Device processing capacity = limited (IoT device)
- Analyze computational cost for both systems
- Estimate which consumes less power and memory
- Justify which cryptosystem is better for the given scenario with reasoning

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method

		strong justification			
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation